Fully automated, Arduino-Based Plant watering System for Smart Agriculture

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***Abstract*—This project presents the design and implementation of an automated plant watering system using Arduino Uno. The system utilizes a soil moisture sensor to monitor the water content in the soil, a DHT11 sensor to measure ambient humidity, and an LM35 sensor for temperature measurement. When the soil moisture drops below a threshold and environmental conditions are suitable, a relay-driven water pump is activated to irrigate the plant. A buzzer provides an audible alert before watering, and a 16x2 LCD displays real-time temperature, humidity, and soil moisture readings. The system aims to minimize manual intervention, conserve water, and ensure healthy plant growth.**

Keywords — Plant watering system; Arduino Uno; Soil moisture sensor; Automated irrigation; Environmental monitoring

I. INTRODUCTION

Manual watering of plants is time-consuming and often leads to under- or over-watering. Automating this process using microcontrollers and sensors can help maintain optimal soil moisture, especially when users are away or busy. This project uses Arduino Uno as the main controller to automate plant irrigation based on real-time sensor data, enhancing plant care efficiency and reliability.

II. MATERIALS REQUIRED

The following components were utilized to build the soil moisture monitoring system:

- Arduino Uno or compatible microcontroller
- Soil Moisture Sensor
- Breadboard and jumper wires
- DHT11 sensor
- LM35 sensor
- USB cable for Arduino-PC connection
- 16x2 LCD display for on-site readings
- Relay module
- 5V Water pump
- Breadboard power supplier
- Buzzer on GPOI board
- Flower pot for testing
- Potentiometer

III. SYSTEM OVERVIEW

A. Soil Moisture Sensor

The capacitive soil moisture sensor measures the volumetric water content of the soil by detecting changes in capacitance. Unlike resistive sensors, capacitive sensors are less prone to corrosion, offering longer durability. The sensor outputs an analog voltage corresponding to the moisture level, which the Arduino reads and processes.

B. Arduino Microcontroller

The Arduino Uno reads the analog signal from the soil moisture sensor through its ADC (Analog-to-Digital Converter). Based on predefined threshold values, it determines the soil's moisture status and provides feedback via serial communication and an LED indicator.

C. DHT11 Sensor

The DHT11 combines a capacitive humidity sensor and a thermistor to measure ambient humidity and temperature. It communicates digitally with the Arduino, providing real-time environmental data.

D. LM35 Sensor

It is a precision analog temperature sensor with a linear output of 10 mV/°C. It provides temperature readings to the Arduino without requiring external calibration.

E. Relay Module

It acts as an electrically operated switch, allowing the Arduino to control high-power devices like the water pump. It isolates the microcontroller from the pump's power supply for safety.

F. Water Pump

It is a DC device that moves water from a reservoir to the plant when activated by the relay. It is powered by an external supply and ensures timely irrigation.

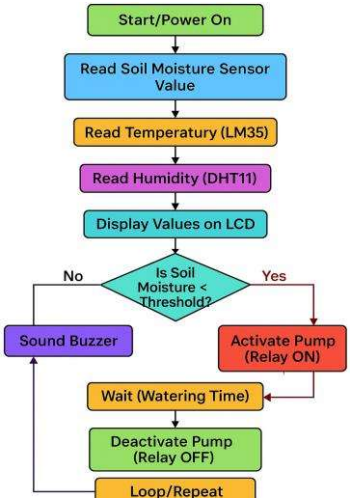

Fig 1: schematic of the project

IV. IMPLEMENTATION STEPS

1. **Sensor Connections:** The automated plant watering system integrates several sensors and actuators with the Arduino Uno to monitor environmental conditions and control irrigation. The soil moisture sensor's VCC and GND pins are connected to the Arduino's 5V and GND, while its analog output is wired to analog pin A0 for moisture readings. The DHT11 sensor is powered by 5V and GND, and its data pin connects to digital pin D7 to provide humidity and ambient temperature data. The LM35 temperature sensor is similarly powered, with its output connected to analog pin A2 for precise temperature measurement. For actuation, the relay module's VCC and GND are connected to the Arduino's 5V and GND, and its control input (IN) is wired to digital pin D6. The water pump's positive terminal is connected to the relay's COM terminal, while the relay's NO terminal is linked to the positive terminal of an external power supply; the pump's negative terminal is connected to the power supply's negative, and all grounds are tied together for a common reference. The 16x2 LCD display is interfaced using six digital pins (D2–D5, D11, D12) and a potentiometer for contrast adjustment. Finally, the buzzer is connected to digital pin D8 and GND, providing an audible alert before watering. This configuration ensures reliable sensor readings, safe relay operation, and effective automated irrigation.

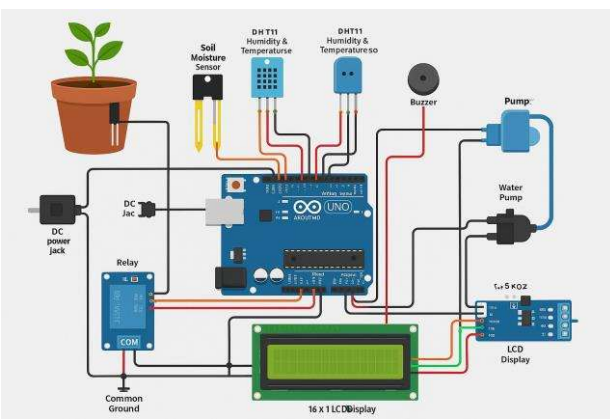

Fig 2: Circuit diagram

2. **Programming:** Upload the code to the Arduino Uno using Arduino IDE.

3. **Testing:** The soil moisture sensor was inserted near the plant's roots for accurate readings, and the water pump was placed in a reservoir with its outlet directed to the plant pot. After powering on the system, real-time temperature, humidity, and soil moisture values were monitored on the LCD. Thresholds in the code were adjusted as needed to match the plant's requirements and environmental conditions.

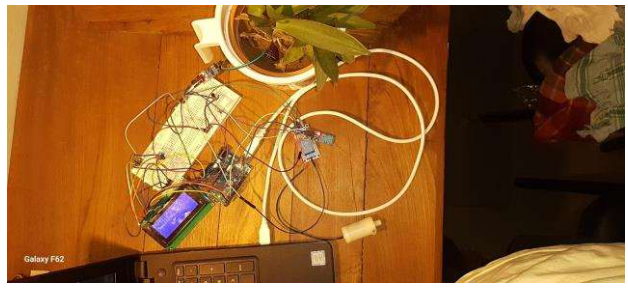

Fig. 3: Photograph of the Arduino based Plant watering system

V. RESULTS AND DISCUSSION

The system effectively measured soil moisture levels, providing real-time feedback through both the LCD display and the serial monitor. The buzzer alert offered an effective warning before watering, while the relay-controlled pump was activated only when necessary, automating the irrigation process efficiently. The use of a capacitive soil moisture sensor, which is resistant to corrosion, ensured consistent long-term performance, though calibration may be required for different soil types and environmental conditions. The moving average filter applied to the LM35 sensor readings resulted in stable and reliable temperature measurements. Throughout testing on various soil types, the system responded accurately to changes in moisture, temperature, and humidity, demonstrating robust and dependable operation for automated plant care.

VI. CONCLUSIONS

The Arduino-based automated plant watering system efficiently manages irrigation by integrating soil moisture, temperature, and humidity sensors. The use of a relay-driven pump, buzzer alert, and LCD interface makes the system user-friendly and reliable. This project demonstrates the potential for low-cost, scalable solutions in smart agriculture and home automation.

VII. ACKNOWLEDGEMENT

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VIII. REFERENCES

[1] Arduino.cc, "Arduino Uno".

[2] Instructables, "Automatic Plant Watering System Using Arduino in an Easy Way".

[3] Fig 1 and Fig 2 both are AI generated.